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Arduino UNO R4 WiFi VRTC & OFF Pins

Learn how to use the VRTC and OFF Pins on the UNO R4 WiFi.

Author · [Jacob Hylén](#) · Last revision · 10/07/2025

The Arduino UNO R4 WiFi features 2 pins that have not been seen before on UNO boards, the VRTC pin and the OFF pin.

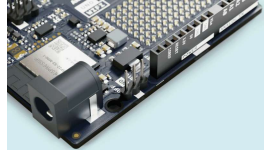
They are used to control some of the boards electrical functions. The VRTC pin can be used to keep the onboard RTC (Real Time Clock) running even when the boards main power supply is turned off, and the OFF pin is used to turn off the board by cutting off the power.

Goals

In this tutorial you will learn how to use the VRTC and the OFF Pins on the Arduino UNO R4 WiFi.

You will learn about how you can use them, why you would use them, and some of the limitations that come with them.

[Help](#)



VRTC and OFF Pin
header

Hardware & Software Needed

Arduino UNO R4 WiFi

A small battery or other
power supply

Jumper cables

VRTC Pin

This guide will not go in
detail on how to use the
RTC feature itself, only
how to use the VRTC Pin.



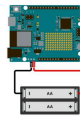
If you're looking for how
to use the RTC features of
the board, check out the

[RTC Guide](#)

The UNO R4 WiFi has a built in
RTC (Real Time Clock) that can
accurately keep track of time.
RTCs are found in many of your
gadgets, although often
connected to them will be a
small battery, to keep the clock
running even when the gadget
is turned off. This is for example
how your laptop knows what
time it is when you start it up,
even if it's disconnected from
the internet.

The UNO R4 WiFi provides the option for you to build a system similar to this in function, by exposing the RTCs power lines so that you can keep it running, even when the boards main power supply is disconnected.

On the header that is located by the barrel jack, you'll find the VRTC pin. And to use it, just apply a voltage within the range of 1.6 - 3.3 V to that pin. This can be done either with a battery pack like shown in the circuit diagram below, but also with other power supplies that slot within the required voltage range.



Battery Pack
Powering the UNO R4
WiFi RTC

The following sketch will start the RTC but only set the time if it this is the first time starting the board since adding the VRTC battery.



```
1 #include "RTC.h"
2
3 void setup() {
4     Serial.begin(9600
5     RTC.begin();
6
7     // A fallback time
8     RTCTime mytime(6,
9
10    // Tries to retri
11    RTCTime savedTime
12    RTC.getTime(saved
13
14
15    if (!RTC.isRunning
16        // this means th
17        if (savedTime.g
18            RTC.setTime(m
19        } else {
20            RTC.setTime(s
21        }
22    }
23 }
24
25 void loop() {
26
27     RTCTime currentti
28     RTC.getTime(currel
```

OFF Pin

The OFF pin on the Arduino UNO R4 WiFi board lets you turn the boards onboard 5 V power supply off, basically turning off the board.

However, it will only turn off the board when it is powered through the VIN pin, or the barrel jack. Why is this? Because by using this pin, you are turning off the step down converter that generates 5 V from whatever voltage you are providing it with. If you are

cable, and there is no need for this step down converter to begin with.

To use the OFF pin, all you need to do is to create a short circuit from it to a GND connection, like in the diagram below. To experiment, you can do this with a jumper cable, but for your finished projects you may want to incorporate a button or switch that will turn the board on or off in this way.



OFF Pin Shorted to
GND

Summary

This short tutorial showed how to use the VRTC and OFF pins that are found on the new header on the Arduino UNO R4 WiFi. These features are brand new to the UNO family.

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